

Minahil Ali

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EDUCATION

National University of Computer and Emerging Sciences (FAST-NU)

Bachelor of Computer Science — Aug 2022 - June 2026

Relevant Courses: Artificial Intelligence, Deep learning, Generative AI, DevOps, Statistical Modelling, Parallel and Distributed Computing, Software Engineering, Compiler Construction, Operating Systems.

A-Levels - 2 A* and 1 A

IGCSE - 9 A*s

EXPERTISE

Technical Skills:

- **Programming Languages:** C++, C, C#, Python, Java, JavaScript, SQL, Bash, Assembly (x86 – MASM)
- **Systems & DevOps:** Docker, Kubernetes, GitHub Actions, AWS, Terraform, Ansible, ArgoCD, Prometheus, Grafana
- **Software & Development Tools:** Git, Docker, Kubernetes, MATLAB, VS Code, Postman, Jupyter, SFML, OpenCL, OpenMP, JavaFX, Jira, Trello
- **Web & APIs:** MERN Stack, Flask, FastAPI, Firebase, Netlify
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Machine Learning & AI:** PyTorch, TensorFlow, Hugging Face, Transformers, CNNs, GANs, Diffusion Models, RNNs/LSTMs, Attention Mechanisms, Multimodal Learning, RAG, Feature Engineering, Prompt Engineering.
- **Tools:** Git, Jupyter, MATLAB, VS Code, Jira, Trello
- **Operating Systems:** Linux, Windows

SELECTED PROJECTS

- **Pose2Play: Vision-Based Pose Estimation for Gamified Rehabilitation Systems (Final Year Project)** – Built a real-time pose estimation system that maps body movements to motion-controlled rehab mini-games, enabling camera-based physical therapy with planned VR integration.
- **Microservices-Based Event Booking Platform** – Built a scalable platform using microservices with Docker, Kubernetes, MongoDB, and PostgreSQL. Integrated Terraform, Ansible, and GitHub Actions for DevOps automation, with ArgoCD for GitOps deployment. Implemented monitoring with Prometheus and Grafana.
- **Multimodal RAG System (Python, LLMs)** – Implemented an end-to-end multimodal RAG pipeline for text and image-based PDFs using OCR, multimodal embeddings, vector search, and advanced prompt engineering for grounded LLM responses.
- **ProjectPulse (MERN Stack)** – Built a task management web application with boards, lists, and sub-tasks. Implemented Redux state management, Tailwind CSS UI, and activity tracking.
- **Synthetic Music Detection (Python, Audio ML)** – Developed an attention-based hybrid model combining CNN, autoencoder, and transformer features to detect AI-generated music, outperforming baselines with strong ROC-AUC and representation separation.
- **Multi-Robot Path Planning (Python, AI)** – Developed an AI-driven pathfinding algorithm using heuristic search and real-time collision avoidance in a dynamic grid.

WORK EXPERIENCE

- **Full Stack Developer Intern** (TechXpert — Jun 2025 – Aug 2025)
 - Delivered multiple production ready MERN applications including a stock trading platform and social media app
 - Integrated AI-powered features using Groq APIs and implemented Firebase based notifications
 - Collaborated in Agile teams (stand-ups, sprint planning) and shipped client facing deployments via Netlify
- **Lab Demonstrator, IICT and Data Structures**
National University of Computer and Emerging Sciences (FAST-NU) — Aug 2024 - Dec 2024
Responsibilities include assisting students during lab sessions, troubleshooting coding issues, conducting practical demonstrations, and providing guidance on assignments and projects related to data structures and IICT concepts.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner (Cloud Foundations)** – Foundational knowledge of AWS core services, cloud architecture principles, security, and billing models.